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Title : JMeter Performance Testing & CI/CD Integration

1) Create the JMeter Test Plan : Create a JMeter test plan containing five Thread Groups.

2) Configure the Thread Groups

Configure each Thread Group with:

- Number of Users: **10**
- Ramp-Up Period: **5 seconds**
- Loop Count: **3**

3) Configure Test Scenarios: Add the required HTTP Request samplers, test data, parameters ,listeners and assertions for each application also

4) Execute the Test in Non-GUI Mode: Execute the JMeter test using the command line to perform the load test efficiently.

5) Analyze Results Using Aggregate Graph/Report

Open the generated JTL file in JMeter and analyze the performance results using the Aggregate Graph/Report.

6) Validate Performance Metrics

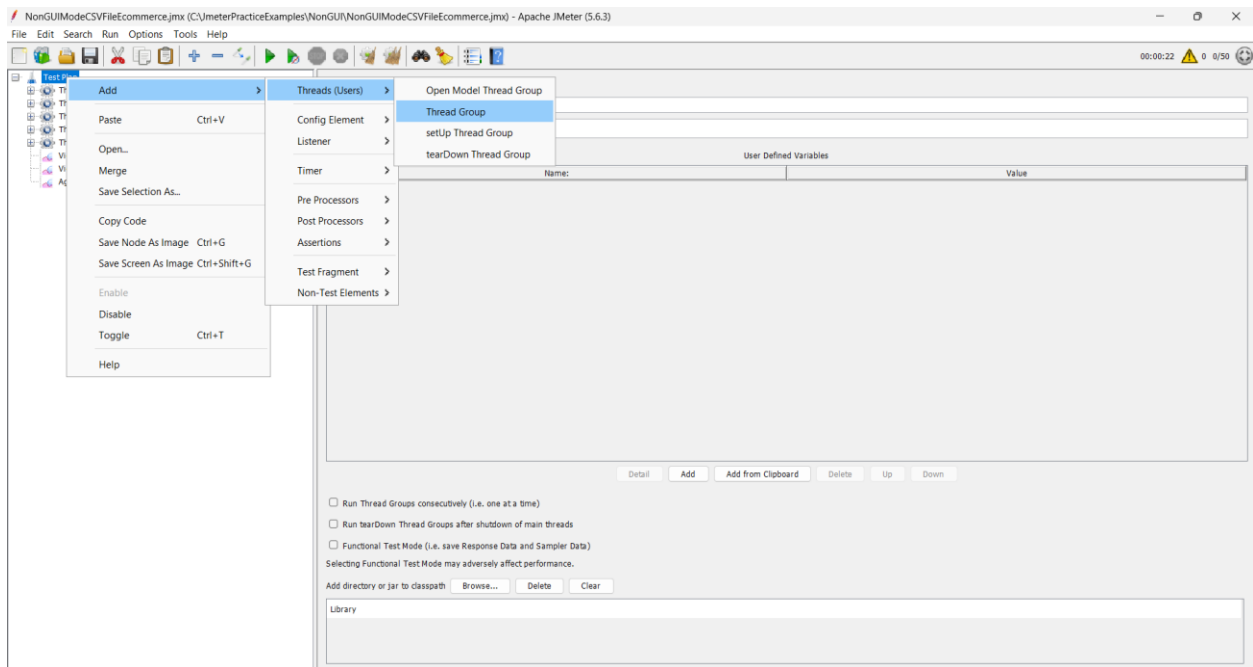
- Total Samples
- Throughput
- Latency
- Median
- Minimum Response Time
- Maximum Response Time
- 90th Percentile
- 95th Percentile
- Error Percentage

7) Generate the HTML Dashboard Report and Integrate with jenkins

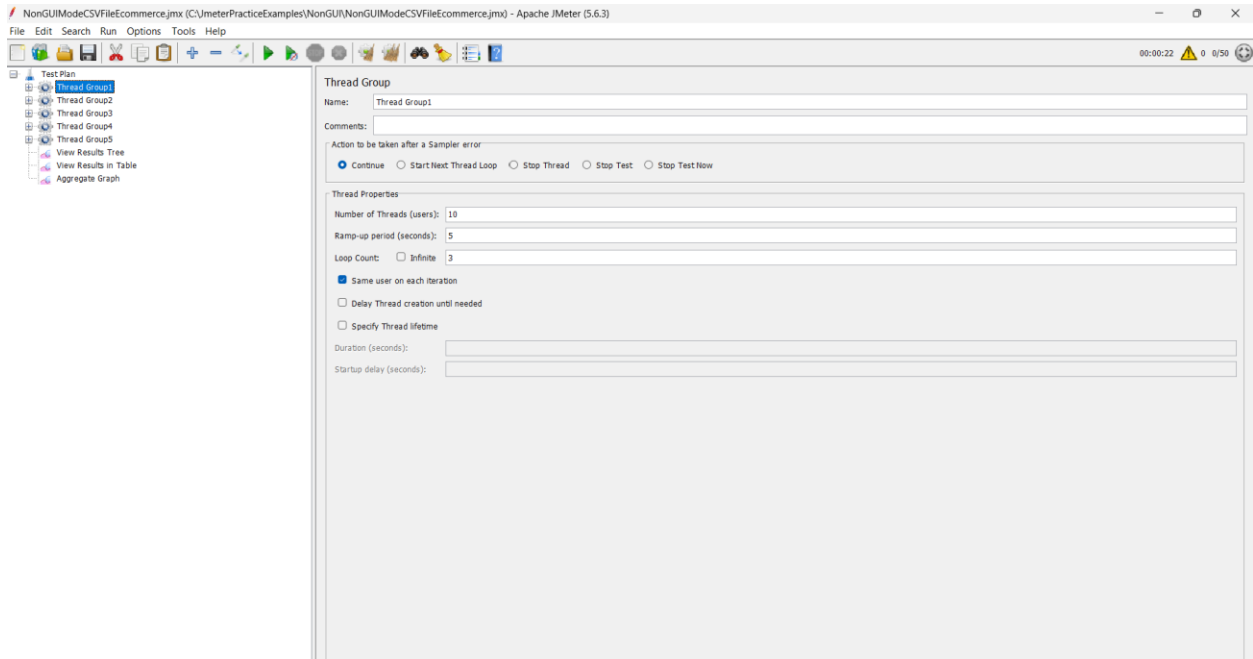
Use the JTL file to generate the JMeter **HTML Dashboard Report** containing graphical performance statistics.

8) Validate Jenkins Execution

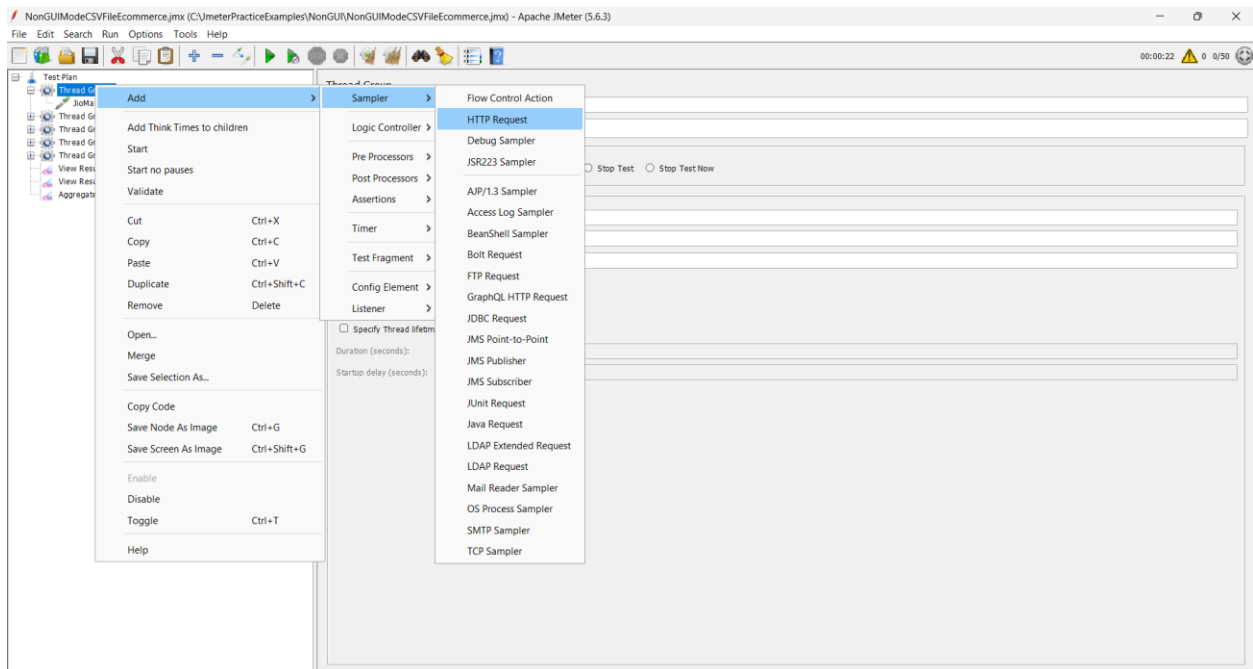
Step 1: Create a JMeter test plan containing five Thread Groups



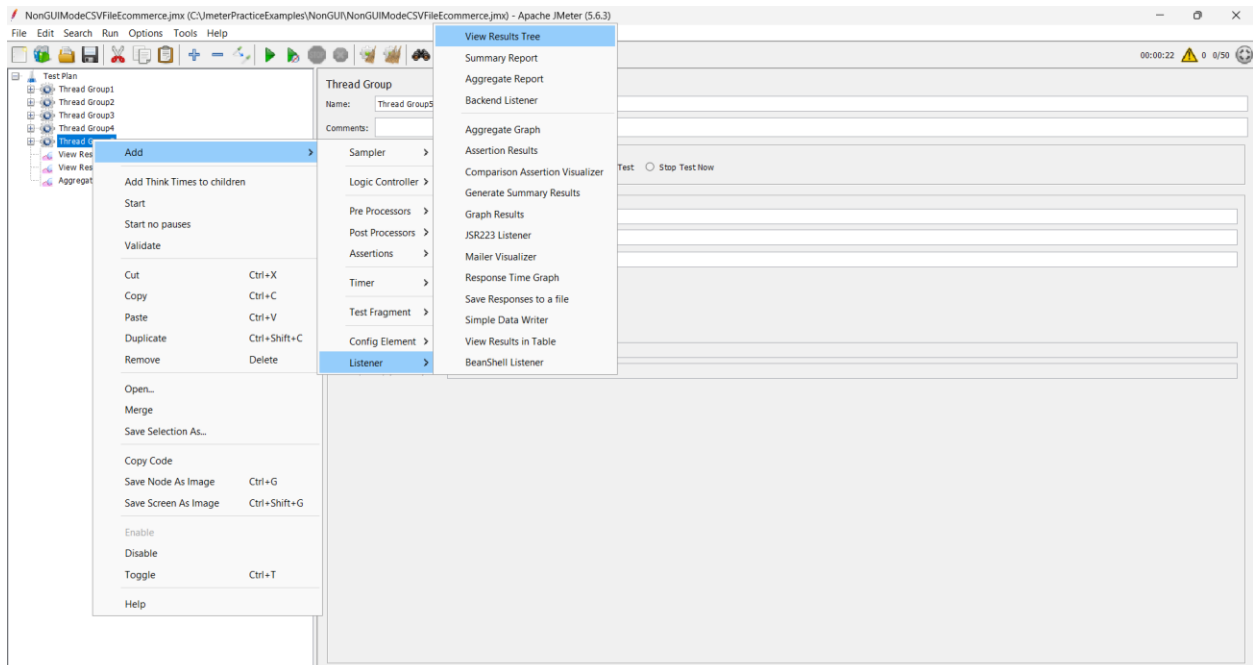
Step 2: Configure each Thread Group according to the requirements mentioned

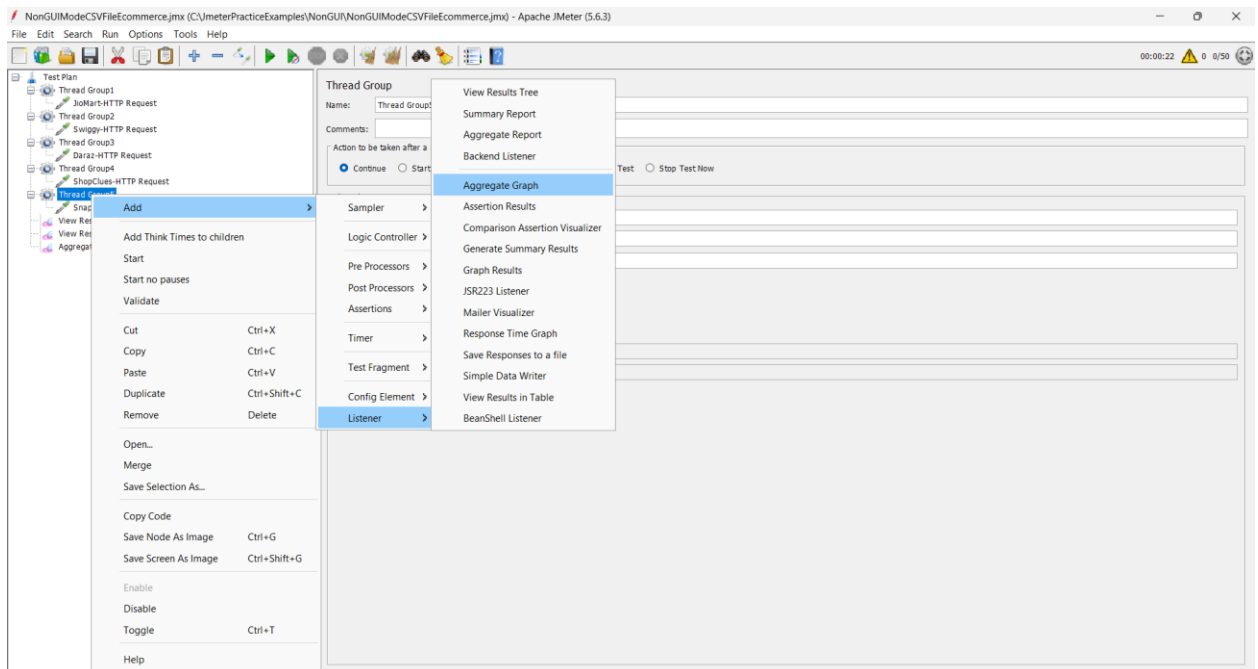
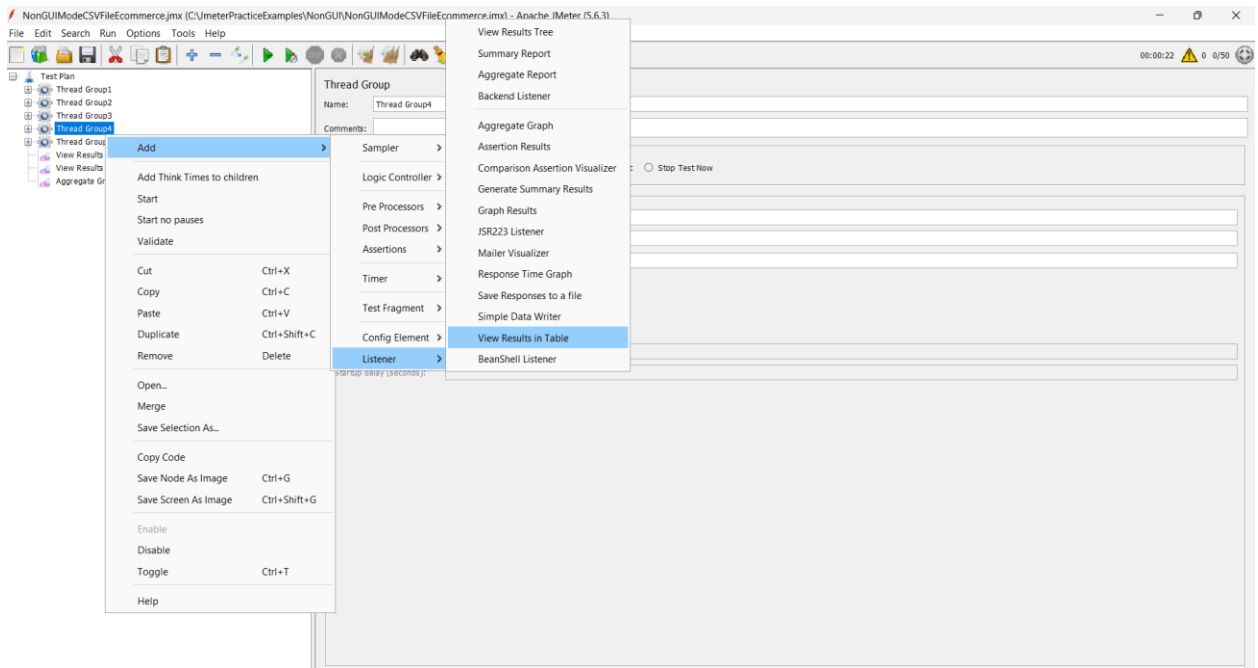


Step 3: Add samplers to each thread group using different E-commerce websites



Step 4: Add listeners





Step 5 : Run and validate

NonGuiModeCS\FireCommerce.jmx (C:\MeterPracticeExamples\NonGui\NonGuiModeCS\FireCommerce.jmx) - Apache JMeter (5.6.3)

File

Edit

Search

Run

Options

Tools

Help

00:00:22 0 /50

Test Plan

Thread Group1

JioMart-HTTP Request

Thread Group2

Suaggy-HTTP Request

Thread Group3

Daraz-HTTP Request

Thread Group4

ShopClues-HTTP Request

Thread Group5

SnapDeal-HTTP Request

View Results Tree

View Results in Table

Aggregate Graph

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename

☐ Log/Display Only
 ☐ Errors
 ☐ Successes

Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
1	18:19:03.410	Thread Group4 4-1	ShopClues-HTTP Req...	789		316269	143	712	669
2	18:19:03.768	Thread Group4 4-2	ShopClues-HTTP Req...	436		316270	143	354	312
3	18:19:04.201	Thread Group4 4-3	ShopClues-HTTP Req...	48		316268	143	19	0
4	18:19:04.250	Thread Group4 4-1	ShopClues-HTTP Req...	49		316269	143	20	0
5	18:19:04.205	Thread Group4 4-2	ShopClues-HTTP Req...	110		316267	143	34	0
6	18:19:04.316	Thread Group4 4-2	ShopClues-HTTP Req...	78		316268	143	25	0
7	18:19:03.410	Thread Group5 5-1	SnapDeal-HTTP Req...	1063		386397	154	1016	647
8	18:19:04.268	Thread Group4 4-3	ShopClues-HTTP Req...	211		316267	143	136	94
9	18:19:04.474	Thread Group5 5-1	SnapDeal-HTTP Req...	106		386391	154	32	0
10	18:19:04.480	Thread Group4 4-3	ShopClues-HTTP Req...	124		316267	143	54	0
11	18:19:04.580	Thread Group5 5-1	SnapDeal-HTTP Req...	111		385921	154	41	0
12	18:19:04.605	Thread Group4 4-3	ShopClues-HTTP Req...	119		316267	143	47	0
13	18:19:03.767	Thread Group5 5-2	SnapDeal-HTTP Req...	1218		386078	154	823	290
14	18:19:04.269	Thread Group5 5-3	SnapDeal-HTTP Req...	734		386115	154	515	35
15	18:19:03.764	Thread Group3 3-2	Daraz-HTTP Request	1335		228574	311	443	346
16	18:19:04.766	Thread Group5 5-4	SnapDeal-HTTP Req...	390		385959	154	162	111
17	18:19:03.410	Thread Group3 3-1	Daraz-HTTP Request	1772		228605	311	798	700
18	18:19:05.003	Thread Group5 5-3	SnapDeal-HTTP Req...	211		385859	154	69	0
19	18:19:05.156	Thread Group5 5-4	SnapDeal-HTTP Req...	216		385967	154	82	0
20	18:19:04.765	Thread Group4 4-4	ShopClues-HTTP Req...	635		316267	143	339	196
21	18:19:05.214	Thread Group5 5-3	SnapDeal-HTTP Req...	234		385967	154	69	0
22	18:19:05.372	Thread Group5 5-4	SnapDeal-HTTP Req...	244		385967	154	101	0
23	18:19:04.985	Thread Group5 5-2	SnapDeal-HTTP Req...	713		385859	154	75	0
24	18:19:05.400	Thread Group4 4-4	ShopClues-HTTP Req...	351		316267	143	134	0
25	18:19:04.266	Thread Group3 3-3	Daraz-HTTP Request	1640		228610	311	302	171
26	18:19:05.751	Thread Group4 4-4	ShopClues-HTTP Req...	271		316268	143	115	0
27	18:19:05.268	Thread Group5 5-5	SnapDeal-HTTP Req...	780		385967	154	240	213
28	18:19:03.761	Thread Group2 2-2	Swiggy-HTTP Request	2395		355313	126	1921	296
29	18:19:06.048	Thread Group5 5-5	SnapDeal-HTTP Req...	329		385967	154	145	0
30	18:19:05.267	Thread Group4 4-5	ShopClues-HTTP Req...	1234		316268	143	581	365
31	18:19:05.099	Thread Group3 3-2	Daraz-HTTP Request	1411		228590	311	169	0
32	18:19:05.183	Thread Group3 3-1	Daraz-HTTP Request	1478		228605	311	183	0
33	18:19:06.378	Thread Group5 5-5	SnapDeal-HTTP Req...	286		385967	154	185	0

☐ Scroll automatically?
 ☐ Child samples?
 No of Samples 150
 Latest Sample 1083
 Average 2924
 Deviation 3018

The screenshot displays the Apache JMeter GUI. The top menu bar includes File, Edit, Search, Run, Options, Tools, and Help. The toolbar contains icons for file operations (Open, Save, Print, etc.) and execution (Start, Stop, etc.). The left sidebar shows the Test Plan tree with the following structure:

- Test Plan
 - Thread Group1
 - JioMart-HTTP Request
 - Thread Group2
 - Swiggy-HTTP Request
 - Thread Group3
 - Daraz-HTTP Request
 - Thread Group4
 - ShopClues-HTTP Request
 - Thread Group5
 - SnapDeal-HTTP Request
 - View Results in Table
 - Aggregate Graph

The main window displays the View Results Tree. The Name field is set to "View Results Tree". The Comments field is empty. The Write results to file / Read from file section shows the Filename field with a Browse button. The Search section includes a search box and buttons for Case sensitive, Regular exp., Search, and Reset. The Test section shows a list of test results for each request, all marked with green checkmarks, indicating successful execution. The results are as follows:

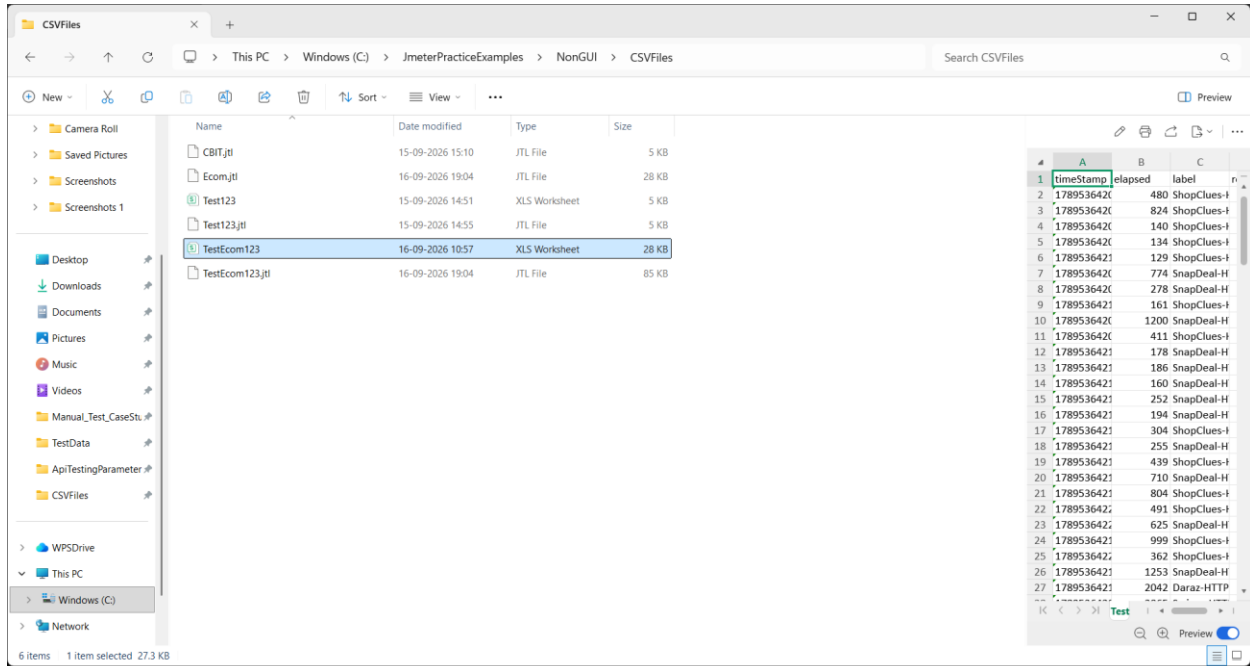
- ShopClues-HTTP Request (10 successful)
- SnapDeal-HTTP Request (10 successful)
- Daraz-HTTP Request (10 successful)
- Swiggy-HTTP Request (10 successful)
- JioMart-HTTP Request (10 successful)

The bottom of the View Results Tree has a checkbox for "Scroll automatically?" and buttons for "Raw" and "Parsed".

Step 6:

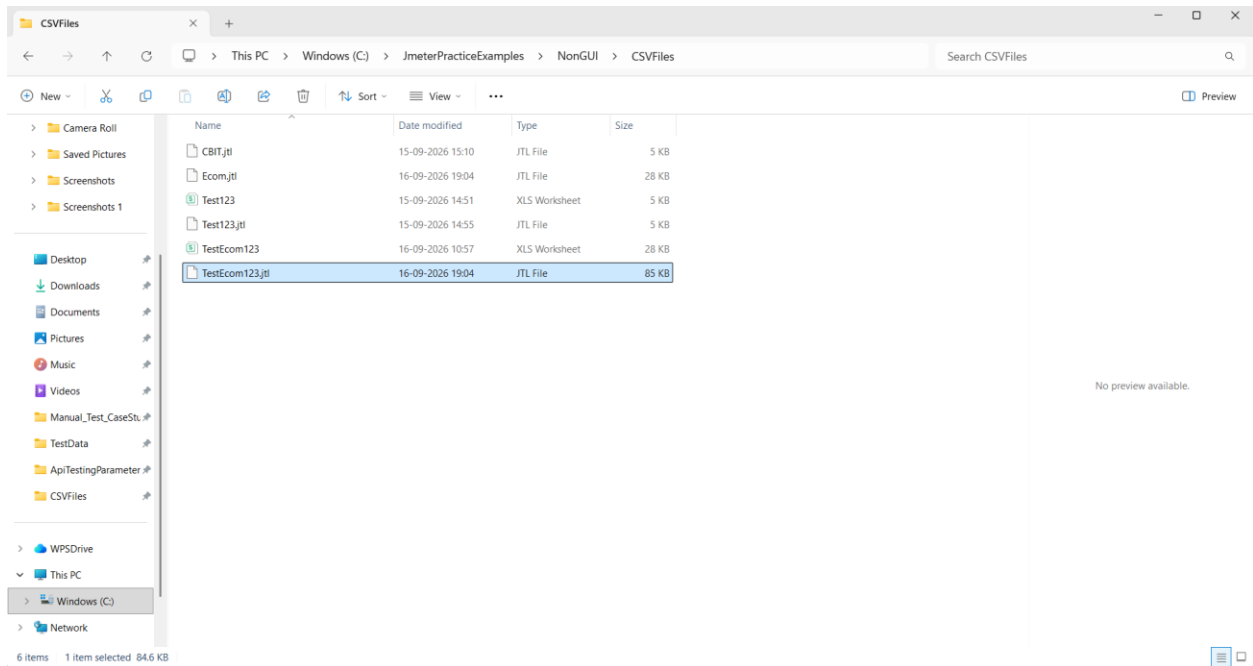
1. Open Command Prompt.
2. Navigate to the JMeter bin folder using the following command:
3. `cd /d "C:\ApacheJmeter5.6.3\apache-jmeter-5.6.3\bin"`
4. Generate the CSV result file by executing the JMeter test plan in non-GUI mode: `jmeter -n -t "C:\JmeterPracticeExamples\NonGUI\NonGUIModeCSVFileEcommerce.jmx" -l "C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.csv"`

The test execution results are saved in the specified location as a CSV file.

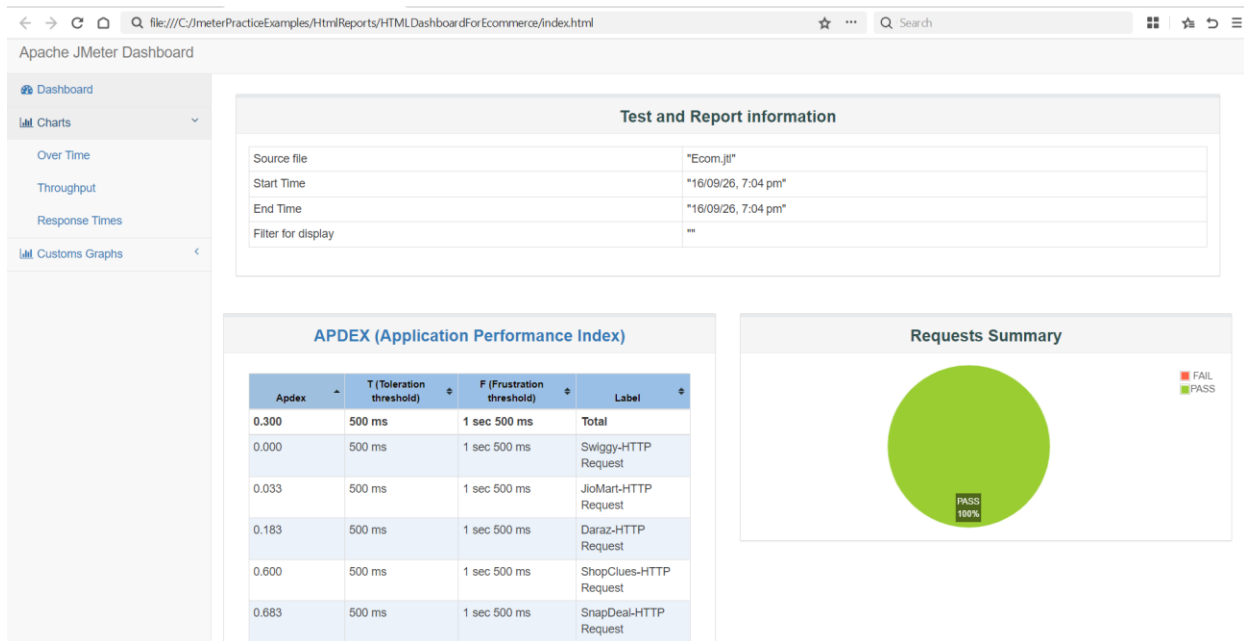
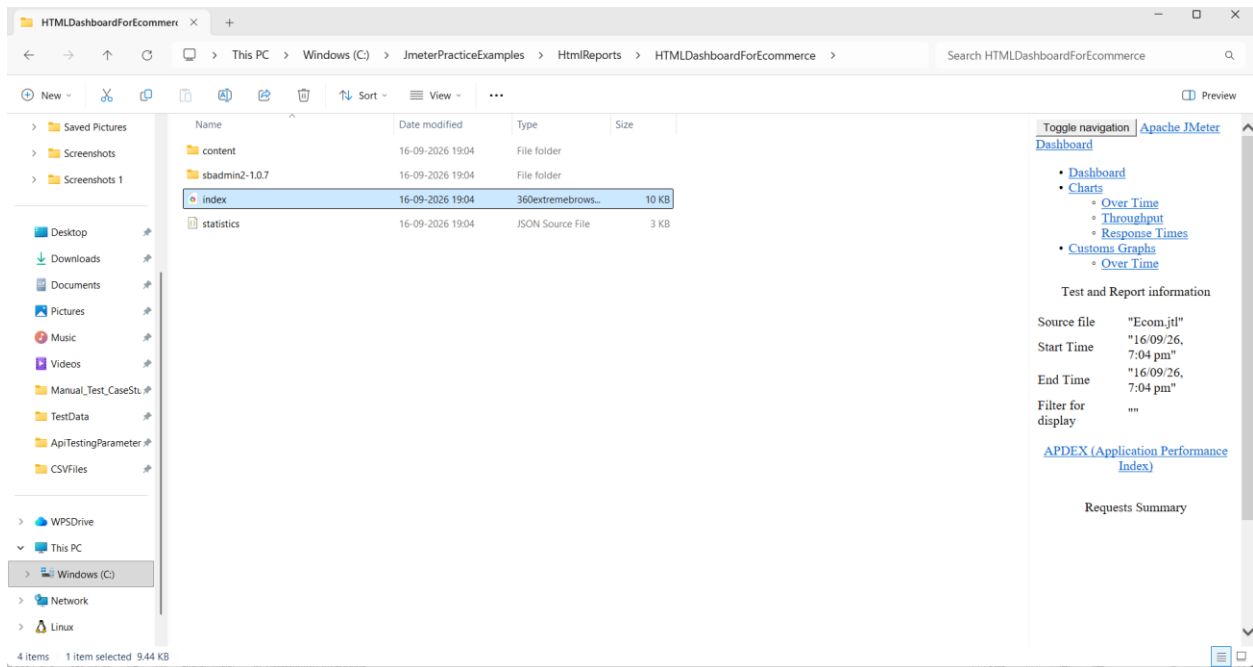


Step 7:

1. Generate the JTL file to use as the input for the Aggregate Graph: `jmeter -n -t "C:\JmeterPracticeExamples\NonGUI\NonGUIModeCSVFileEcommerce.jmx" -l "C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.jtl"`
2. Open the generated JTL file in JMeter and use the Aggregate Graph/Report to analyze the performance results, including Throughput, Latency, Median, Minimum, Maximum, Percentiles, Total Samples, and Error Percentage.



Step 7: Html reports



Apache JMeter Dashboard

Apache JMeter Dashboard

file:///C:/JmeterPracticeExamples/HtmlReports/HTMLDashboardForEcommerce/Index.html

Search

Statistics

Requests	Executions				Response Times (ms)							Throughput	Network (KB/sec)	
	Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	150	0	0.00%	2736.43	38	15742	1847.00	5502.50	7396.10	15671.62	6.51		10476.60	1.11
Daraz-HTTP Request	30	0	0.00%	2150.57	653	5836	1736.00	4740.30	5625.35	5836.00	2.33		521.23	0.71
JioMart-HTTP Request	30	0	0.00%	5120.67	1025	15742	3522.00	15381.70	15666.10	15742.00	1.30		8839.84	0.18
ShopClues-HTTP Request	30	0	0.00%	997.53	46	3224	835.50	2588.10	3224.00	3224.00	2.90		896.40	0.41
SnapDeal-HTTP Request	30	0	0.00%	623.97	38	2045	559.00	1540.30	1935.00	2045.00	3.66		1380.50	0.55
Swiggy-HTTP Request	30	0	0.00%	4789.43	2290	7995	4391.50	7184.60	7798.10	7995.00	1.59		551.69	0.20

Step 8: Run through Jenkins – create freestyle project

Jenkins / All / New Item

New Item

Enter an item name

JenkinsContinuousIntegrationContinuousDeploymentUsingJmeter

Select an item type

- Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- ☒ Maven project
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder
Creates a container that groups related items in it. Useful for organizing things together. Hello view

OK

Step 9 : Configure Jenkins

 Jenkins

JenkinsCICDUsingJmeter

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command

See the list of available environment variables

cd C:\ApacheJmeter5.6.3\apache-jmeter-5.6.3\bin
jmeter -Tjmeter.save.saveservice.output_format=xml -n -t
"C:\JmeterPracticeExamples\CICDPracticeTests\NonGUIModeCSVFileEcommerce.jmx"
-l "C:\JmeterPracticeExamples\CICDPracticeTests\JenkinsCICD.jtl"

Advanced

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Publish Performance test result report

Save

Apply

Step 10: Analyze the performance Trend

 Jenkins

JenkinsCICDUsingJmeter

Performance Trend

Status

Changes

Workspace

Build Now

Delete Project

Performance Trend

Configure

Rename

Test Results Analyzer

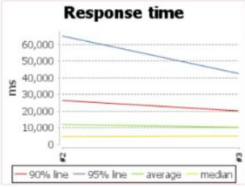
Performance Trend

Last Report

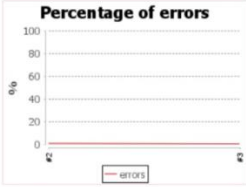
Filter trend data

Test file: JenkinsCICD.jtl

Response time



Percentage of errors



Trend report

Builds >

Filter

Today

#3 1:54 pm

#2 11:47 am

#1 11:44 am



Trend report: JenkinsCICD.jtl

Build	Line 90.0(ms)	Line 95.0(ms)	Max(ms)	Median(ms)	Min(ms)	Percentage of errors	average	average kb	errors	total kb
#2	26267	65067	179643	4495	357	1	11833	1606.88	1	241031.39
#3	19997	42468	179643	4923	228	0	10139	1608.68	1	482603.23

Jenkins 2.568.2

Step 11: JMeter Aggregate Graph – Performance Test Results

